UNIVERSITY OF LIVERPOOL

Big Data Week 8 Lab

Knowledge Graph Builder

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Honorary Professor

November 11, 2025



Today’s Lecture

Week 8 Lab: Knowledge Graph Builder

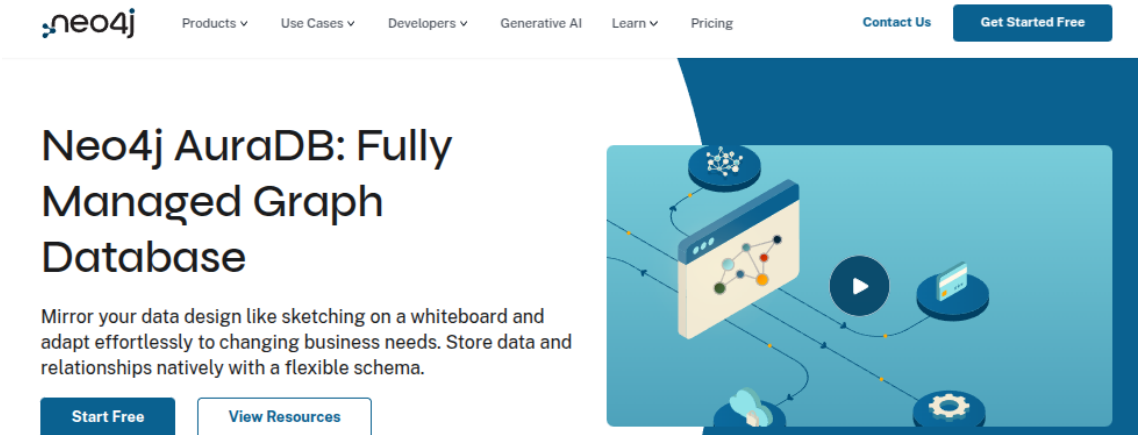


One of 8,000 **Starlink satellites** in orbit, beaming high-speed internet to the most remote corners of the Earth.

Week 8 Lab: Knowledge Graph Builder

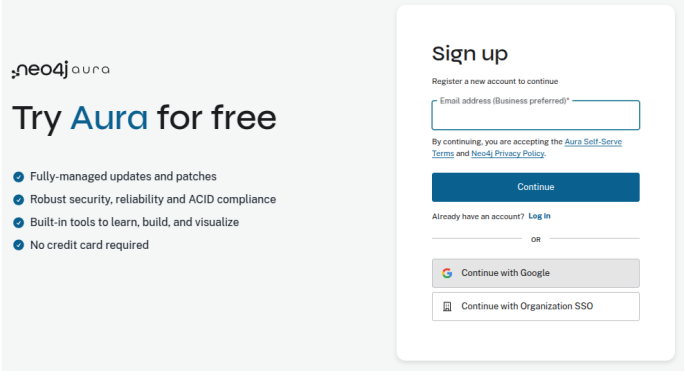
Neo4j Website

- Navigating to the Neo4j **website**, you can see an invitation to “Get Started Free” on the top right.
- Clicking on this will take you to the sign up page for Aura. . .



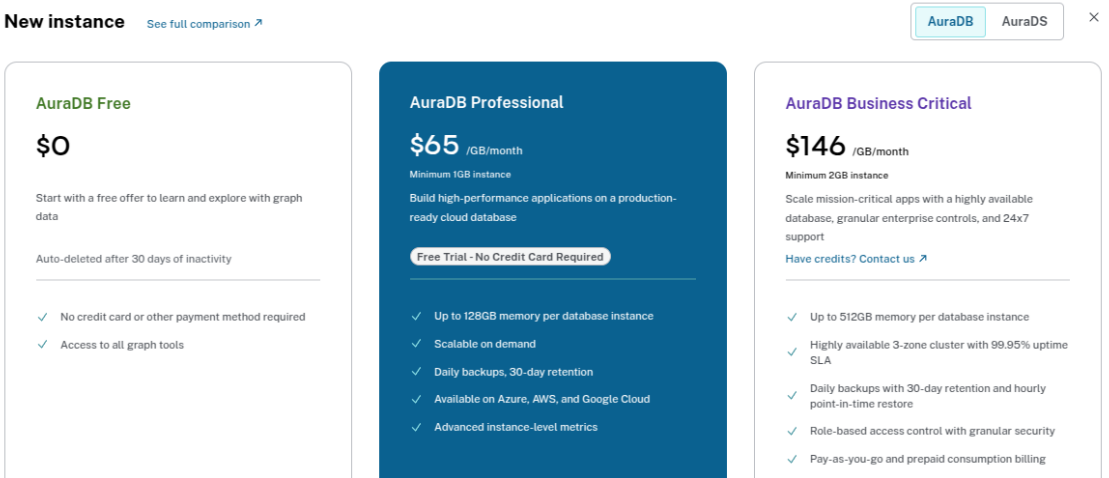
Aura Sign up

- On the Aura **sign up page**, you can create a free account.



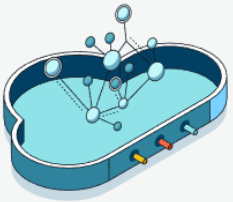
Aura Sign up Options

- Select the “AuraDB Free” option.



Aura Instance Creation

- You can then create a database:



Create your first instance

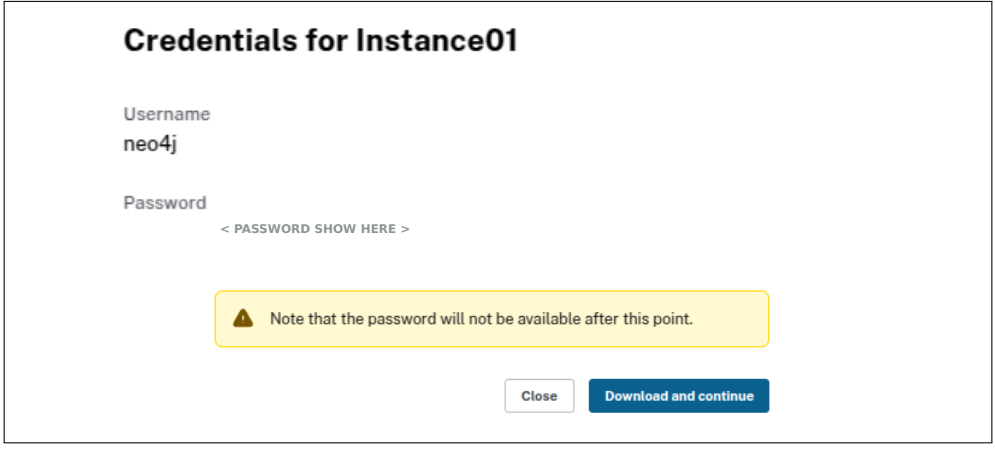
Configure your instance and start realizing the endless possibilities with graphs.

Create instance

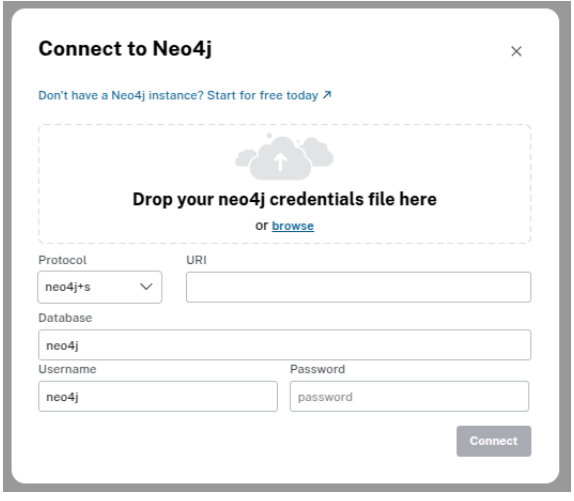
Have data? Import existing data into Neo4j.

Neo4J Credentials

- At this point, your login credentials will be shown
- **IMPORTANT**: Download these, because you will need them to connect to the database, and they'll never be shown to you again.



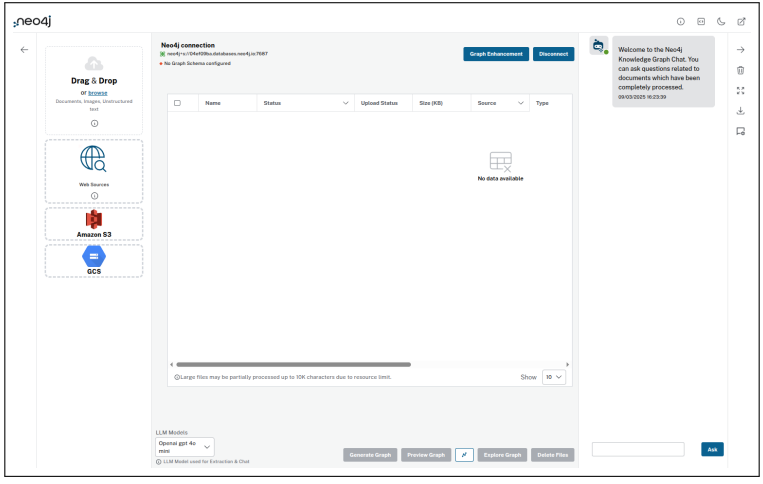
Connect to Neo4j Database



Creating the Knowledge Graph

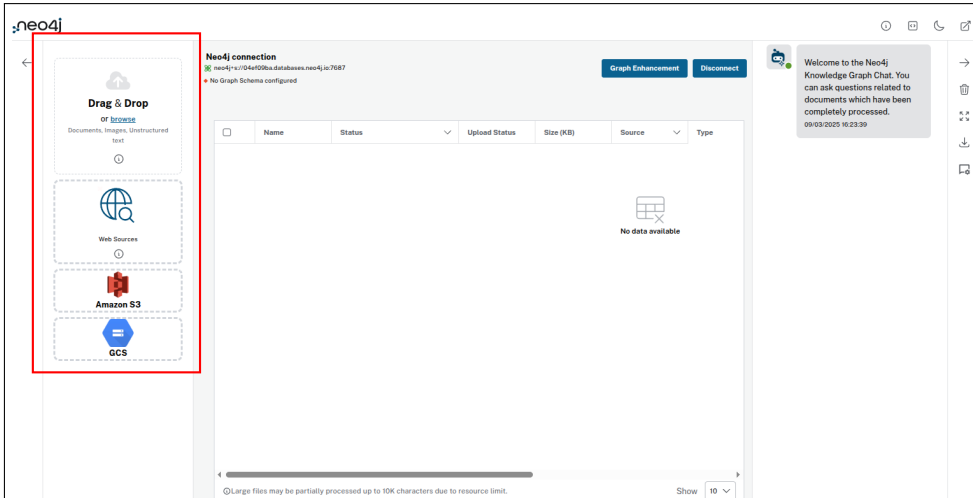
- You can find the LLM Graph Builder **here**.
- The process starts by ingesting your unstructured data.
- An LLM is then used to identify key entities and their relationships.
- You can upload data using three input modes:
  - Drag and drop PDFs and other files into the first input zone.
  - Copy/paste a YouTube video link in the second input.
  - Enter a Wikipedia page link in the third input.
- We'll now run through an example to demonstrate how this works.
- We'll use some data relating to COP 27...

Knowledge Graph Builder



Uploading Data

Uploading is done with this panel:



Uploading Wikipedia Data

Here's the Wikipedia page for the 2022 United Nations Climate Change Conference.



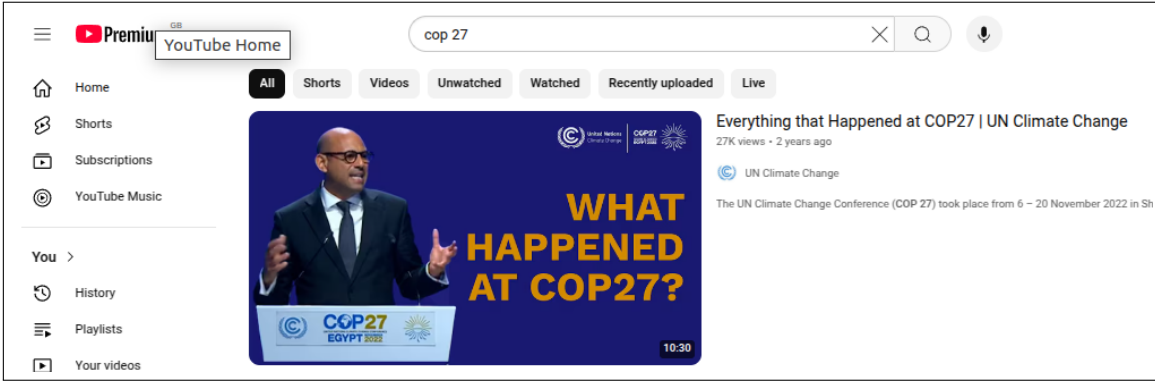
Uploading YouTube Videos

We can upload Wikipedia pages to the Graph Builder as follows:



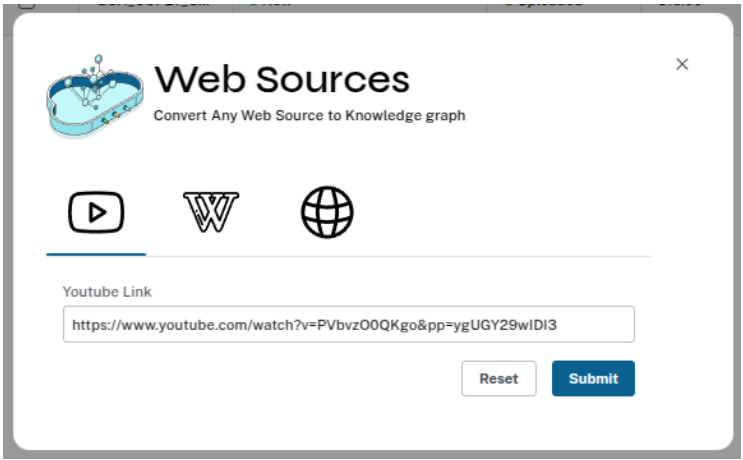
Uploading YouTube Videos

Here's a YouTube video summarising COP 27.



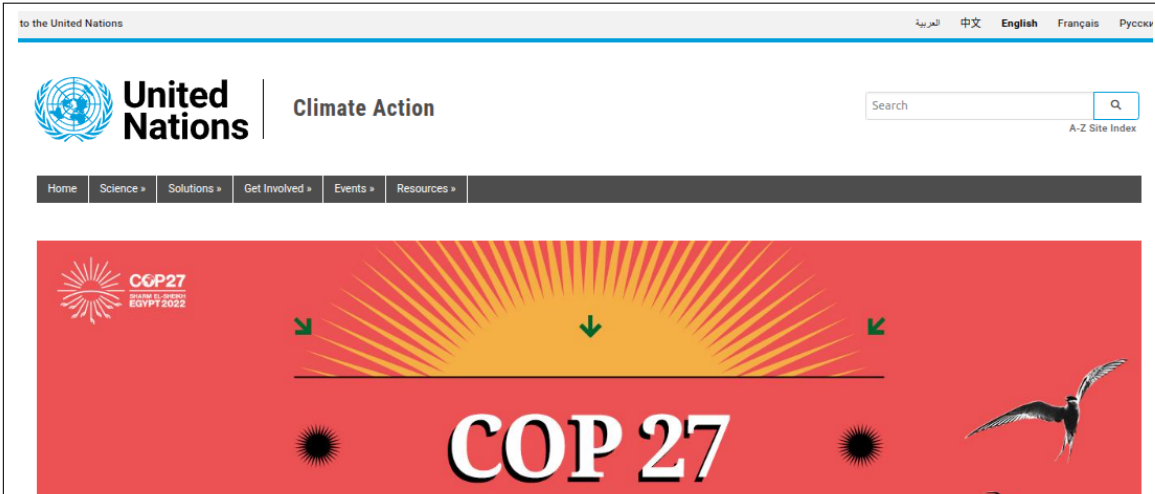
Uploading YouTube Videos

We can upload YouTube videos to the Graph Builder as follows:



Uploading Webpages

Here's the UN webpage for COP 27.



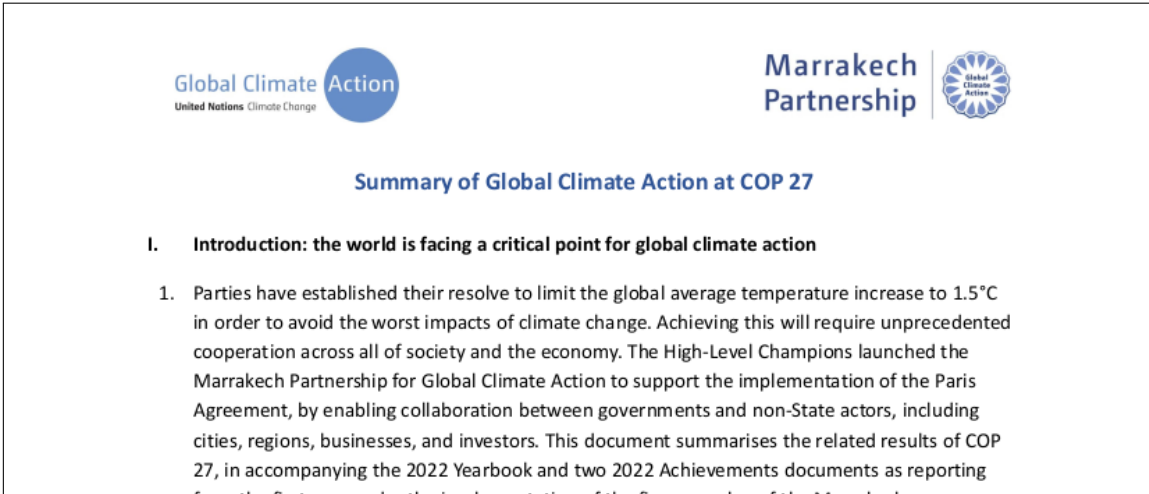
Uploading YouTube Videos

We can upload YouTube videos to the Graph Builder as follows:



Uploading PDFs

Here's a PDF summarising COP 27:



Global Climate Action
United Nations Climate Change

Marrakech Partnership

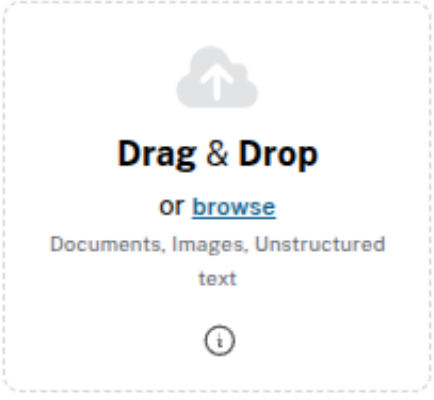
Summary of Global Climate Action at COP 27

I. Introduction: the world is facing a critical point for global climate action

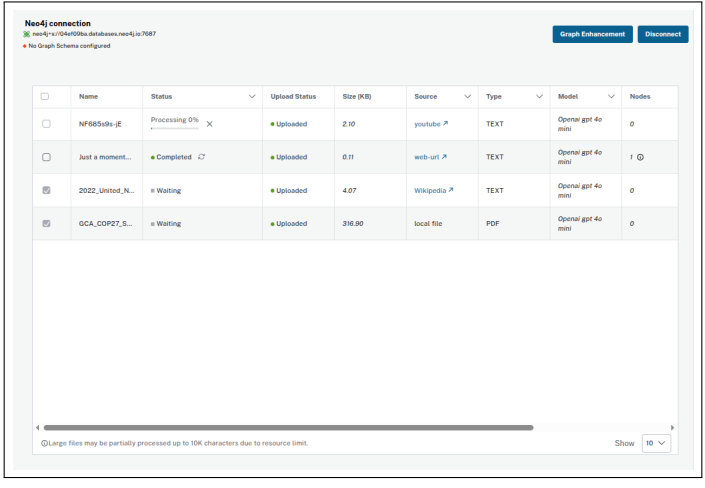
1. Parties have established their resolve to limit the global average temperature increase to 1.5°C in order to avoid the worst impacts of climate change. Achieving this will require unprecedented cooperation across all of society and the economy. The High-Level Champions launched the Marrakech Partnership for Global Climate Action to support the implementation of the Paris Agreement, by enabling collaboration between governments and non-State actors, including cities, regions, businesses, and investors. This document summarises the related results of COP 27, in accompanying the 2022 Yearbook and two 2022 Achievements documents as reporting

Uploading PDFs

PDFs can be uploaded to the Graph Builder as follows:



Knowledge Graph Builder with Data

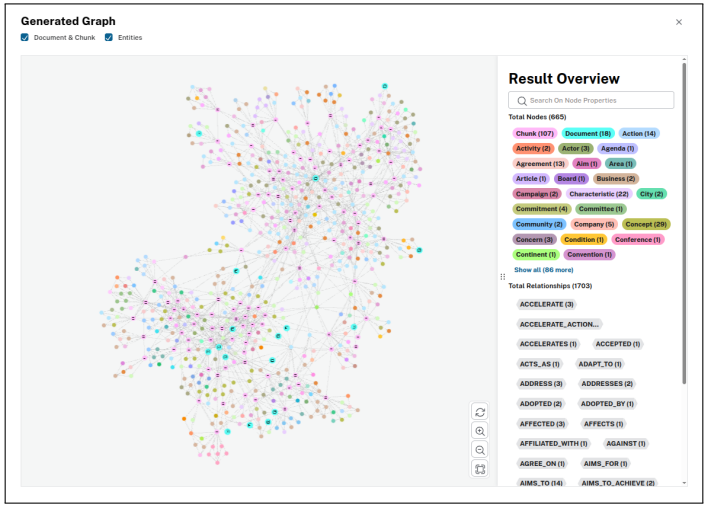


Knowledge Graph Preview

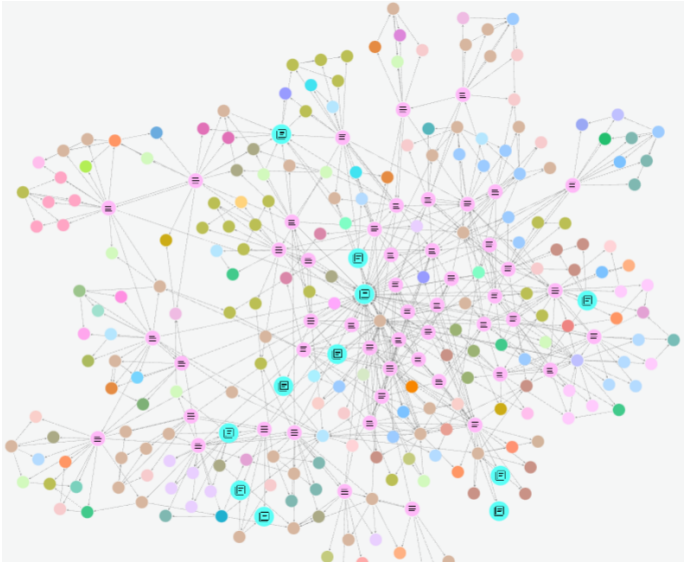
- Once all files are uploaded, your knowledge graph is automatically constructed.
- The graph visually represents the extracted entities and their relationships.
- This visualisation helps you understand the structure and connections within your ingested data.

Knowledge Graph Preview

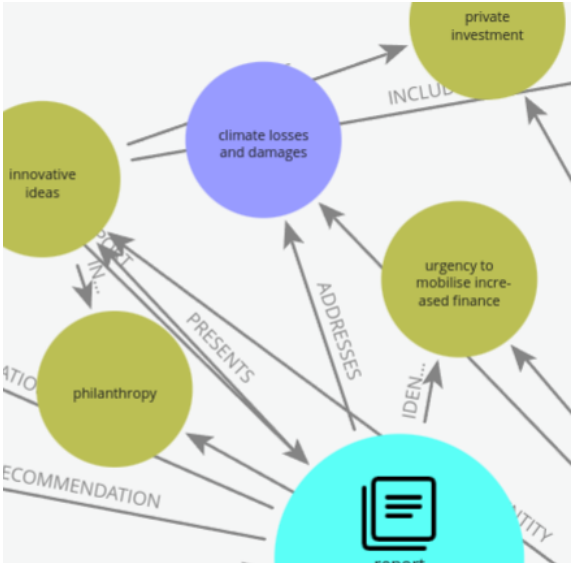
Select some of the documents and click the blue “preview graph” button:



Graph for UN Summary PDF



Zoomed In Graph for UN Summary PDF



Exploring the Graph

Question

Split into 3-5 person groups. In 15 minutes:

- Pick an interesting node in the graph and explore its relationships.
- What insights can you gain from this exploration?
- Are you able to find a research topic from it?

Questions?



Liverpool three graces from the docks by Kevin Jump, CC BY 2.0

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